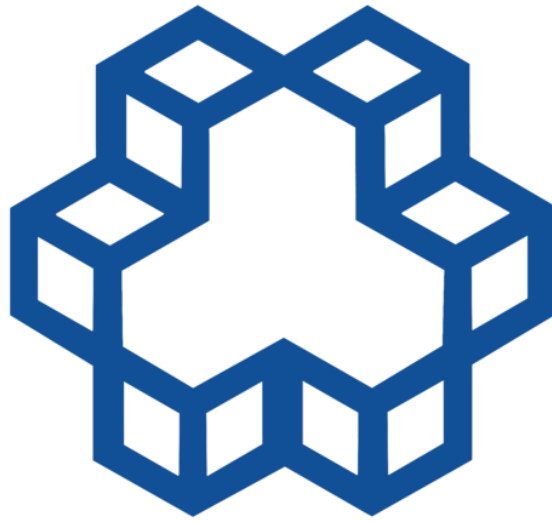




K. N. Toosi University of Technology
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Computer vision and Robotics

Final Project Report: **Plant Health engine**

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1 Introduction

The tomato (*Solanum lycopersicum*) is one of the most widely cultivated and economically significant vegetable crops worldwide. However, its yield and quality are highly susceptible to a variety of foliar diseases caused by fungi, bacteria, and viruses. Traditionally, diagnosing these diseases requires visual inspection by agricultural experts—a process that is not only labor-intensive and time-consuming but also prone to subjective error. As agriculture increasingly transitions toward automated monitoring and the integration of plant digital twin technologies, the demand for rapid, accurate, and computationally lightweight diagnostic systems has become paramount.

In recent years, deep learning—particularly Convolutional Neural Networks (CNNs)—has demonstrated exceptional efficacy in computer vision tasks, including plant disease classification. Despite these advancements, the performance of CNNs remains heavily reliant on the quality of the input data. Variations in illumination, shadow artifacts, and background noise can severely degrade feature extraction and model generalization.

To address these challenges, this study proposes a robust deep learning pipeline for tomato plant disease detection. The methodology introduces a stringent preprocessing phase encompassing automated leaf segmentation to isolate the region of interest, followed by Contrast Limited Adaptive Histogram Equalization (CLAHE) to mitigate lighting variations and enhance the textural features of pathological lesions. Furthermore, this research conducts a rigorous comparative analysis of three state-of-the-art architectures: MobileNetV2, ResNet50, and EfficientNetB0. By systematically evaluating, eliminating, and fine-tuning these models, this study identifies the optimal architecture that balances high predictive accuracy with the computational efficiency required for deployment in resource-constrained environments.

2 Dataset Description

The data utilized in this study is derived from the publicly available PlantVillage dataset, a benchmark repository widely used for image-based plant disease diagnosis. For the scope of this project, the dataset was strictly filtered to include only images of tomato leaves.

The tomato subset comprises thousands of high-resolution RGB images captured in controlled laboratory settings. Each image typically features a single, detached leaf positioned against a relatively uniform background, which facilitates the evaluation of automated segmentation algorithms.

The dataset is categorized into 10 distinct classes, representing one healthy baseline and nine prevalent disease states. The classes are distributed as follows:

- **Healthy:** Leaves exhibiting no visible signs of pathogen infection or nutritional deficiency.
- **Bacterial Spot:** Characterized by small, water-soaked spots that eventually turn black.
- **Early Blight:** Identified by brown or black spots with concentric rings (target-like appearance).
- **Late Blight:** Marked by large, irregularly shaped water-soaked lesions, often leading to rapid tissue necrosis.
- **Leaf Mold:** Presents as pale greenish-yellow spots on the upper leaf surface with corresponding olive-green mold on the underside.
- **Septoria Leaf Spot:** Small, circular spots with dark borders and lighter gray or tan centers.
- **Target Spot:** Brown lesions with distinct concentric rings, occasionally with a yellow halo.
- **Spider Mites (Two-Spotted Spider Mite):** Manifests as a stippled, yellowing appearance on the leaves due to pest feeding.
- **Tomato Mosaic Virus:** Causes a distinct mottling or mosaic pattern of light and dark green on the foliage.
- **Tomato Yellow Leaf Curl Virus:** Characterized by severe yellowing, upward curling of leaf margins, and stunted leaf growth.

Prior to model training, the images were standardized to a uniform resolution to satisfy the input requirements of the selected CNN architectures.

3 Data Acquisition and Exploratory Analysis

The computational pipeline for this study was structured to ensure high data integrity and efficient processing. Following data acquisition, an automated verification protocol was executed to parse the directory structure and validate file formats, ensuring that only compatible, uncorrupted images progressed to the preprocessing phase.

A comprehensive Exploratory Data Analysis (EDA) was conducted to quantify the exact class distribution across the dataset. The algorithm extracted the total volume of valid images and mapped the specific frequency of each of the ten categories (nine diseases and one healthy baseline). Analyzing this distribution is a vital prerequisite in deep learning applications; identifying potential class imbalances early informs subsequent training strategies, ensuring the neural network does not become statistically biased toward majority classes.

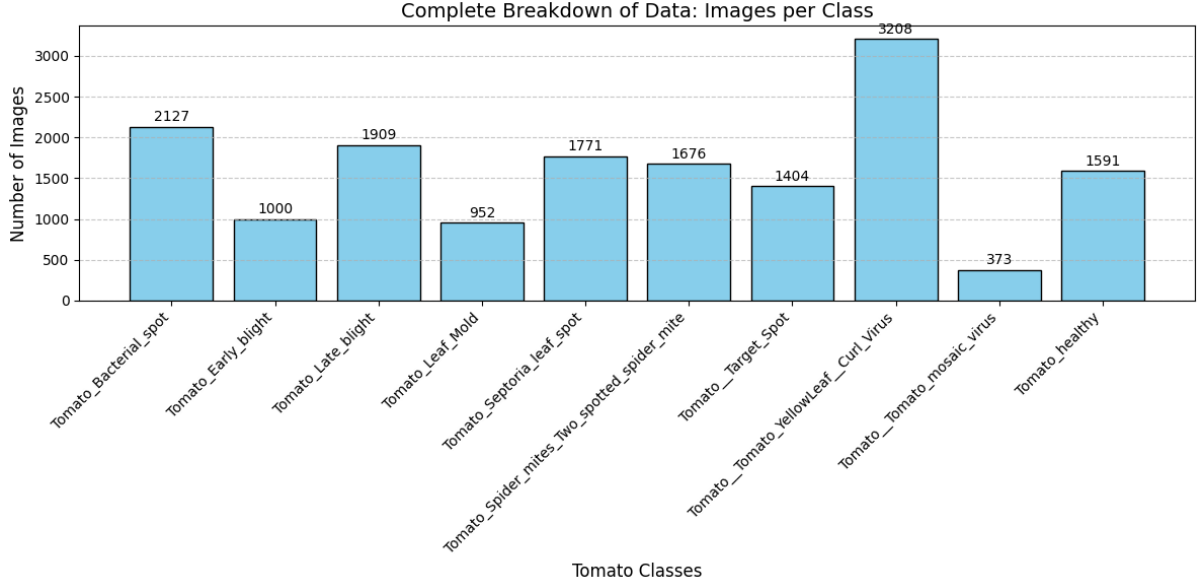


Figure 1: Distribution of tomato leaf images across the dataset classes, quantifying the available training data per category.

In tandem with the quantitative analysis, a qualitative visual inspection was integrated into the pipeline. A randomized sampling function was utilized to extract and visualize image grids across all subdirectories. This step served as a manual verification checkpoint to confirm that the images were correctly routed from their respective label directories and that the visual data maintained high fidelity prior to the application of mathematical image transformations.

3.1 Image Preprocessing and Background Segmentation

While the PlantVillage dataset is captured under controlled conditions, the raw images still contain background variations, shadows, and non-leaf artifacts. If left unprocessed, convolutional neural networks (CNNs) may inadvertently learn these background features as confounding variables, reducing the model’s ability to generalize to real-world agricultural environments. To ensure the models focus exclusively on leaf morphology and pathogenic lesions, a rigorous background subtraction pipeline was implemented.

For salient object detection and extraction, the U-2-Net deep learning architecture was deployed. U-2-Net is a two-level nested U-structure that is highly effective at capturing both local and global contextual information without significantly increasing computational cost, making it ideal for accurately delineating the complex and often irregular boundaries of plant leaves.

To process the large volume of high-resolution images efficiently, the inference pipeline was executed using the ONNX (Open Neural Network Exchange) runtime, explicitly

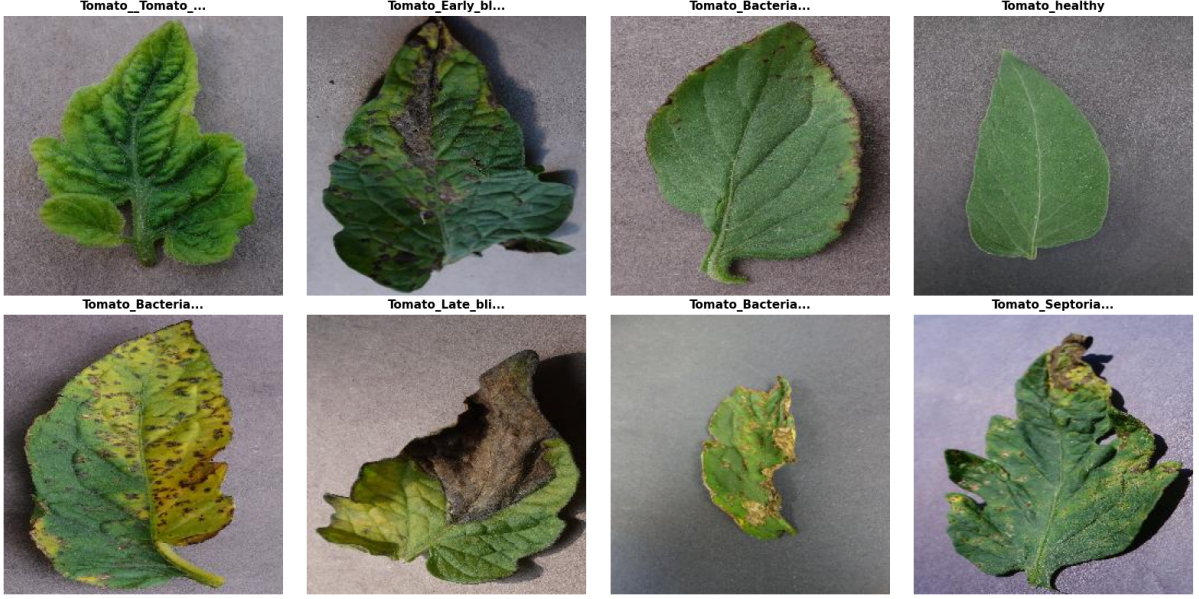


Figure 2: Randomly sampled dataset images utilized for qualitative visual verification of class labels and data integrity.

configured with a CUDA Execution Provider for GPU acceleration. Furthermore, an asynchronous multi-threading architecture was developed to prevent computational bottlenecks. Disk I/O operations—reading raw images and writing processed arrays—were decoupled from the GPU-bound inference tasks, ensuring the GPU remained continuously saturated.

During processing, each image was parsed and passed through the U-2-Net model, which generated a high-precision alpha mask isolating the leaf. This alpha mask was mathematically applied to the original RGB channels. The background space, corresponding to an alpha value of zero, was substituted with a uniform black background ($R = 0, G = 0, B = 0$). This zero-padding approach effectively nullifies the background, guaranteeing that no extraneous gradient updates occur from non-plant pixels during the CNN training phase.

As a final data curation step, an automated integrity check was integrated into the pipeline to identify and purge any corrupted or unreadable image files, ensuring a pristine data corpus for subsequent feature enhancement.

3.2 Feature Enhancement via Contrast Limited Adaptive Histogram Equalization (CLAHE)

Following background segmentation, the dataset underwent localized feature enhancement to amplify the visual morphological symptoms of the diseases, such as necrotic lesions, chlorotic halos, and fungal structures. Standard global histogram equalization is often unsuitable for agricultural imagery, as it tends to over-amplify noise and wash out fine regional details. To overcome this, Contrast Limited Adaptive Histogram Equalization (CLAHE) was employed.

To preserve the integrity of the segmented background, a rigorous masking protocol was integrated into the enhancement pipeline. The operational sequence was defined as follows:

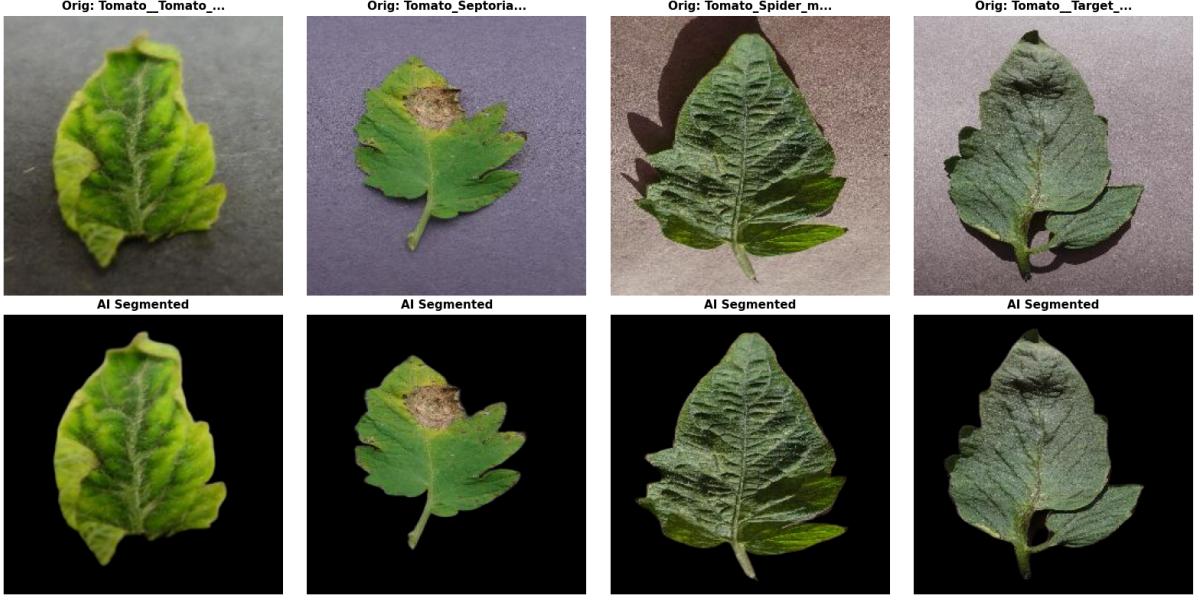


Figure 3: Comparison of original raw images and their corresponding outputs after U-2-Net background segmentation.

First, a binary mask of the isolated leaf was generated by thresholding the grayscale representation of the image, identifying all pixels with a strictly positive intensity value. The original image was then transformed from the standard RGB color space into the CIELAB ($L^*a^*b^*$) color space. This transformation is critical as it decouples the luminance (L^*) from the chromatic components (a^* and b^*).

CLAHE was applied exclusively to the luminance channel (L^*) using a predefined clip limit of 2.0 and a localized tile grid size of 8×8 . By restricting the enhancement strictly to the L^* channel, the algorithm increased the contrast of the pathological textures without introducing artificial color distortion.

Once enhanced, the modified luminance channel was merged with the untouched chromatic channels, and the matrix was converted back to the RGB color space. As a final protective measure, the previously generated binary mask was re-applied using a bitwise AND operation. This guaranteed that the background pixels remained at absolute zero intensity, effectively preventing any background artifact amplification caused by the histogram stretching.

3.3 Dataset Preparation and Class Balancing

Following image preprocessing, the dataset was partitioned into mutually exclusive training, validation, and testing subsets utilizing TensorFlow’s optimized data pipeline API. To satisfy the strict input tensor requirements of the selected convolutional architectures (MobileNetV2, ResNet50, and EfficientNetB0), all images were standardized to a spatial resolution of 224×224 pixels across three color channels ($224 \times 224 \times 3$).

During the exploratory data analysis, a distinct quantitative disparity in the number of samples per class was identified. Training deep learning models on imbalanced datasets frequently induces a predictive bias, wherein the network over-optimizes for majority classes at the expense of underrepresented disease categories. To mathematically mitigate this bias without artificially generating synthetic data, a class weighting strategy was integrated directly into the training loop.

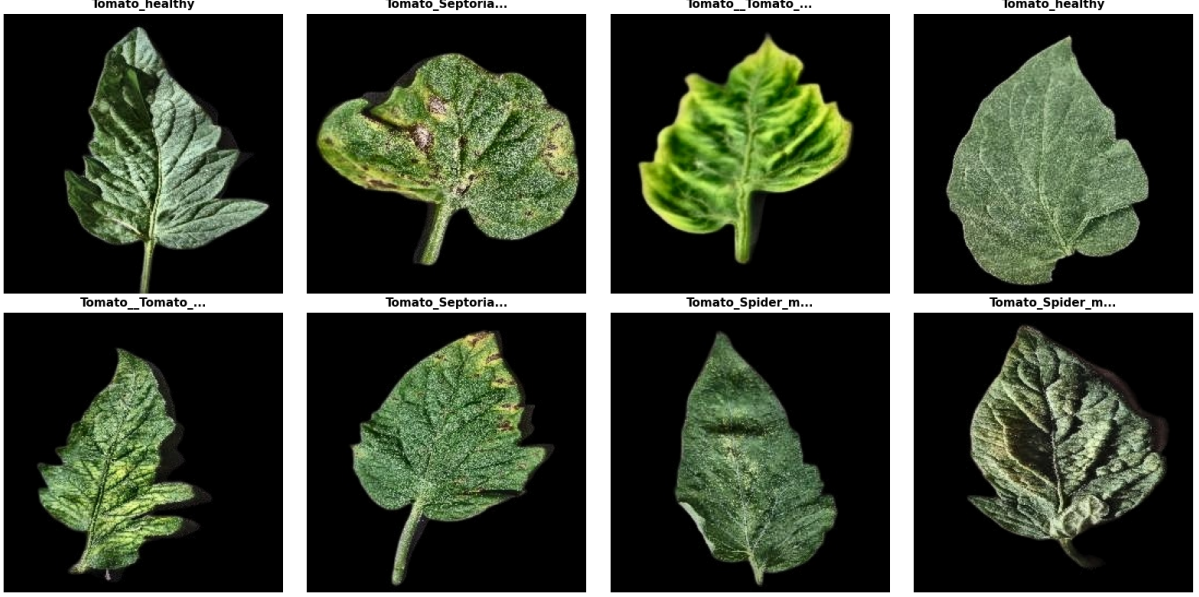


Figure 4: Randomly sampled images post-CLAHE application, demonstrating enhanced lesion contrast and precisely maintained zero-value backgrounds.

During the loss calculation phase, gradient updates were scaled dynamically. The penalty for misclassifying an image was inversely proportional to the frequency of its ground-truth class within the dataset. The specific weight W_i for each class i was computed utilizing the following formulation:

$$W_i = \frac{1}{N_i} \times \frac{N_{total}}{K} \quad (1)$$

where N_i represents the exact number of validated images belonging to class i , N_{total} is the aggregate volume of images across the entire dataset, and K denotes the total number of discrete classes ($K = 10$).

To ensure computational efficiency and prevent memory bottlenecking during back-propagation, the partitioned datasets were processed as dynamically loaded tensor batches rather than being statically stored in RAM. The exact cardinality of the splits was algorithmically verified to guarantee strict data isolation between the training, validation, and testing environments.

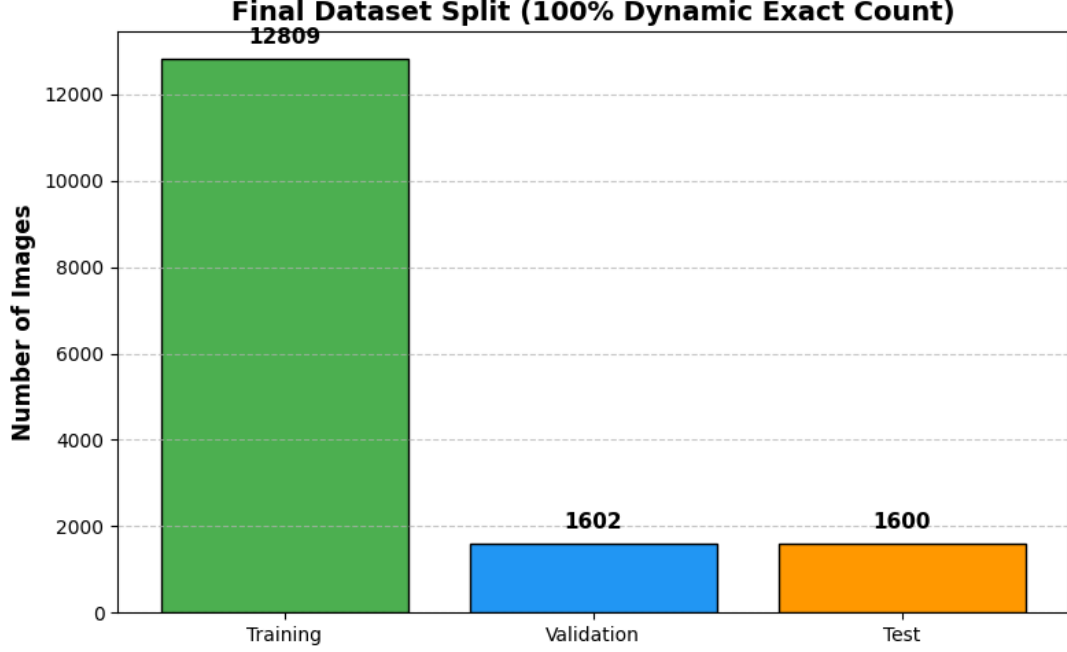


Figure 5: Distribution of the botanical image dataset into training, validation, and testing subsets, processed dynamically as tensor batches.

4 Model Architectures and Initial Training Phase

To identify the most effective neural network for potential edge-device deployment within a plant digital twin framework, a comparative analysis of three prominent lightweight convolutional architectures was conducted: MobileNetV2, ResNet50, and EfficientNetB0. These models were selected due to their proven balance between parameter efficiency and deep feature extraction capabilities.

A transfer learning paradigm was employed to accelerate convergence and leverage highly robust hierarchical feature representations. The base architectures were instantiated with pre-trained weights derived from the ImageNet dataset. To preserve these learned spatial hierarchies during the initial training phase, the base convolutional layers of all three models were strictly frozen (*trainable = False*). Consequently, the models functioned as static feature extractors.

Because each base architecture expects distinct input distributions, model-specific mathematical preprocessing layers were integrated directly into the computational graphs. MobileNetV2 utilized continuous rescaling to map pixel intensities to a $[-1, 1]$ range, while ResNet50 employed its native channel-wise zero-centering transformation. EfficientNetB0 absorbed these requirements internally. Notably, standard dynamic data augmentation pipelines (e.g., random rotations, shearing) were intentionally bypassed to preserve the exact spatial and textural integrity achieved during the prior CLAHE enhancement phase.

Following the frozen base models, a custom classification head was constructed. Global Average Pooling (GAP) was applied to reduce the spatial dimensions of the feature maps to a one-dimensional vector, mitigating the risk of overfitting by significantly reducing the parameter count. A Dropout layer with a rate of 0.2 was subsequently introduced for further regularization. The terminal layer consisted of a fully connected Dense network containing 10 units, utilizing a softmax activation function to output a normalized probability distribution across the target disease classes. The models were compiled using the

Adam optimization algorithm and a sparse categorical crossentropy loss function.

To maximize computational throughput during training, the data pipeline was rigorously optimized. The training and validation datasets were cached directly to disk and asynchronously prefetched into memory utilizing dynamic hardware autotuning. By decoupling the input/output operations from model execution, the GPU remained continuously saturated.

An initial training phase was executed for a duration of 10 epochs. To maintain system stability during this sequential comparative evaluation, the backend computational session was programmatically cleared, and memory was explicitly garbage-collected between each model initialization, preventing VRAM fragmentation. An automated checkpointing algorithm monitored the validation accuracy, ensuring that only the optimal mathematical weights were committed to non-volatile storage, effectively securing the best model state.

5 Initial Results

5.1 Initial Model Evaluation and Transient Response

Following the initial phase of transfer learning with frozen base layers, a comparative analysis of the network learning dynamics was conducted. The primary objective was to evaluate the generalization trajectory of each architecture on the enhanced dataset and identify the most promising models for deeper optimization.

To quantify this, the validation accuracy and sparse categorical cross-entropy loss were evaluated and recorded dynamically at the conclusion of each epoch. The resulting performance curves graphically illustrate the convergence behaviors of MobileNetV2, EfficientNetB0, and ResNet50 over the designated 10-epoch timeframe.

By analyzing the comparative transient response, distinct architectural capabilities emerged. While MobileNetV2 is highly parameter-efficient, empirical observations of the learning curves indicated a comparatively lower convergence ceiling in validation accuracy and a higher stabilization plateau in validation loss relative to its peers. Based on this performance deficit in the initial feature extraction phase, MobileNetV2 was eliminated from the candidate pool.

Conversely, both ResNet50 and EfficientNetB0 exhibited superior generalization metrics, demonstrating rapid accuracy gains and steady loss minimization. Consequently, these two architectures were selected to advance to the network fine-tuning phase.

6 Network Fine-Tuning and Optimization

Following the initial evaluation, both EfficientNetB0 and ResNet50 advanced to a rigorous fine-tuning phase to specialize their hierarchical feature representations for botanical anomaly detection. While the initial training phase utilized pre-trained ImageNet weights strictly for generic feature extraction, fine-tuning allows the deeper layers of the network to adapt to the specific visual domain of the PlantVillage dataset.

The methodology detailed herein highlights the precise optimization protocol applied to EfficientNetB0. To achieve domain adaptation without inducing catastrophic forgetting, a partial unfreezing strategy was implemented. The EfficientNetB0 base architecture was reconstructed and initialized with the optimal weights secured during the first phase. Subsequently, the top 30 convolutional layers of the network were unfrozen (*trainable* =

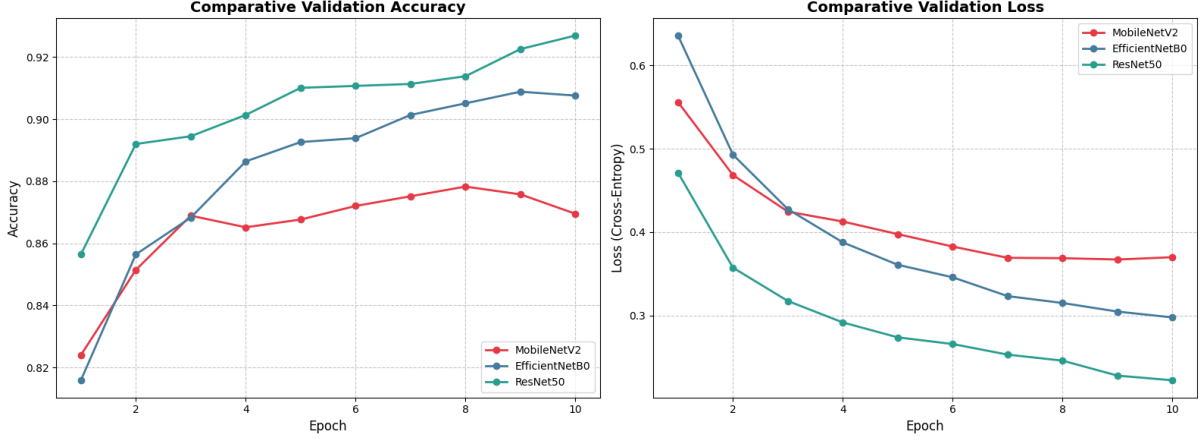


Figure 6: Initial Results plots: showing both ResNet50 and EfficientNetB0 exhibited superior generalization metrics

True), while the earlier layers—responsible for foundational edge, color, and shape detection—remained firmly locked. (A parallel architectural unfreezing strategy was also executed for ResNet50).

The optimization hyperparameters were stringently recalibrated to support this delicate weight update process. The Adam optimizer was retained, but the learning rate was drastically reduced to 1×10^{-5} . This micro-learning rate ensures that gradient updates remain small, gently steering the top-layer weights toward the target domain without destabilizing the robust, pre-existing spatial hierarchies.

Furthermore, to maximize generalization and computational efficiency, a robust suite of dynamic callbacks was integrated into the extended 30-epoch training loop:

- **Model Checkpointing:** Continuously monitored validation accuracy, recording the optimal mathematical state strictly when empirical performance peaked.
- **Adaptive Learning Rate Decay (ReduceLROnPlateau):** Dynamically reduced the learning rate by a factor of 0.2 if validation accuracy stagnated for two consecutive epochs. This allowed the optimizer to descend deeper into the loss minimum without overshooting.
- **Early Stopping:** Preemptively terminated the training process if no improvement was observed over four epochs. This mechanism prevented over-parameterization on the training set and automatically restored the network to its highest-performing epoch.

Following the initial evaluation phase, both EfficientNetB0 and ResNet50 demonstrated sufficient baseline accuracy (approximately 91.2% and 92.5%, respectively) to warrant deeper optimization. To specialize their hierarchical feature representations for botanical anomaly detection, both architectures advanced to a rigorous fine-tuning phase.

7 Final Model Selection and Test Evaluation

Following the fine-tuning phase of both ResNet50 and EfficientNetB0, a final comparative evaluation was conducted using the strictly isolated test dataset. This ensured an unbiased assessment of model generalization on entirely unseen data.

While both architectures demonstrated exceptional diagnostic capabilities, EfficientNetB0 yielded a slightly higher overall test accuracy. Furthermore, EfficientNetB0 achieved this performance utilizing significantly fewer trainable parameters than ResNet50. This superior balance of high predictive accuracy and low computational overhead makes EfficientNetB0 the optimal choice for deployment within resource-constrained environments, such as embedded edge devices in greenhouse digital twins. Consequently, EfficientNetB0 was selected as the final model for this study.

To rigorously quantify the diagnostic performance of the fine-tuned EfficientNetB0 architecture, a comprehensive evaluation suite was executed. Beyond overall accuracy and categorical cross-entropy loss, a detailed class-wise analysis was performed to identify any localized classification deficiencies.

A confusion matrix was generated to visualize the true positive rates against misclassifications, providing insight into which specific disease states the network found visually analogous. From this matrix, standard classification metrics including Precision, Recall, and the F1-Score were derived for each of the 10 botanical categories.

The final fine-tuned EfficientNetB0 model achieved an outstanding overall test accuracy of **95.94%** with a corresponding test loss of **0.1189**. The model demonstrated exceptional robustness across nearly all categories, achieving 100% accuracy for the Tomato Mosaic Virus. The detailed per-class diagnostic performance is summarized in Table 1.

Table 1: Detailed Classification Report and Per-Class Accuracy for EfficientNetB0

Disease Class	Precision	Recall	F1-Score	Accuracy (%)
Healthy	0.97	0.95	0.96	95.38
Bacterial Spot	0.95	0.99	0.97	99.07
Early Blight	0.91	0.88	0.89	87.78
Late Blight	0.99	0.98	0.99	97.78
Leaf Mold	0.98	0.98	0.98	97.56
Septoria Leaf Spot	0.93	0.93	0.93	93.02
Target Spot	0.88	0.92	0.90	91.95
Spider Mites	0.95	0.97	0.96	96.88
Tomato Mosaic Virus	1.00	1.00	1.00	100.00
Yellow Leaf Curl Virus	1.00	0.97	0.99	97.38
Macro Average	0.96	0.96	0.96	-

7.1 Comparative Evaluation and Final Model Selection

Following the fine-tuning phase, a rigorous comparative evaluation of both ResNet50 and EfficientNetB0 was conducted using the strictly isolated test dataset. This parallel evaluation ensured an unbiased assessment of model generalization on entirely unseen data.

To quantify the diagnostic performance of both architectures, a comprehensive evaluation suite was executed. Beyond overall accuracy and categorical cross-entropy loss, detailed class-wise analyses—including Precision, Recall, and F1-Scores derived from localized confusion matrices—were performed to identify any specific classification deficiencies.

The fine-tuned ResNet50 model achieved a robust overall test accuracy of **94.44%** with a corresponding test loss of **0.1922**. While highly effective, it was ultimately out-

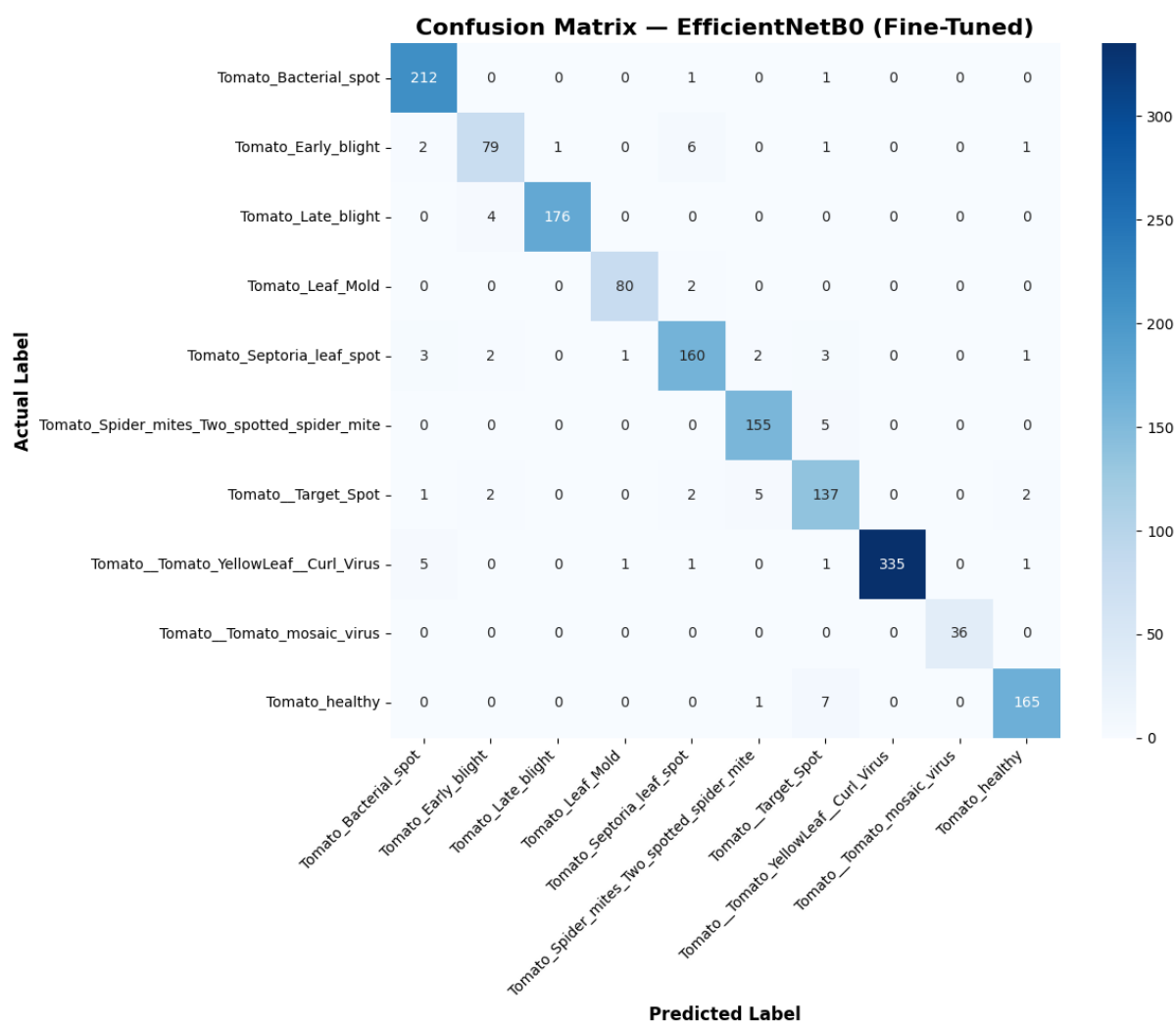


Figure 7: Confusion Matrix for the fine-tuned EfficientNetB0 model on the unseen test dataset, detailing correct predictions along the diagonal and specific misclassification overlaps.

performed by the fine-tuned EfficientNetB0 architecture, which achieved an outstanding overall test accuracy of **95.94%** and a lower test loss of **0.1189**. Furthermore, a class-wise comparison revealed that EfficientNetB0 exhibited superior sensitivity to challenging morphological variations; for instance, it achieved 100% accuracy on the Tomato Mosaic Virus, whereas ResNet50 recorded 94.44% on the same class.

Beyond raw predictive performance, architectural efficiency was a primary selection criterion for this study. The target deployment environment imposes strict constraints on computational resources and memory overhead. EfficientNetB0 achieved superior diagnostic accuracy while utilizing significantly fewer trainable parameters than the heavier ResNet50 architecture.

This optimal intersection of high diagnostic precision and low computational overhead formally justified the selection of EfficientNetB0 as the final deployed model for the plant digital twin module. A comparative summary of the overarching performance metrics is detailed in Table 2.

Table 2: Final Comparative Performance of Fine-Tuned Architectures on the Unseen Test Set

Architecture	Test Accuracy	Test Loss	Parameters (Approx.)	Selection Status
ResNet50	94.44%	0.1922	~ 23.5 Million	Eliminated
EfficientNetB0	95.94%	0.1189	~ 4.0 Million	Selected

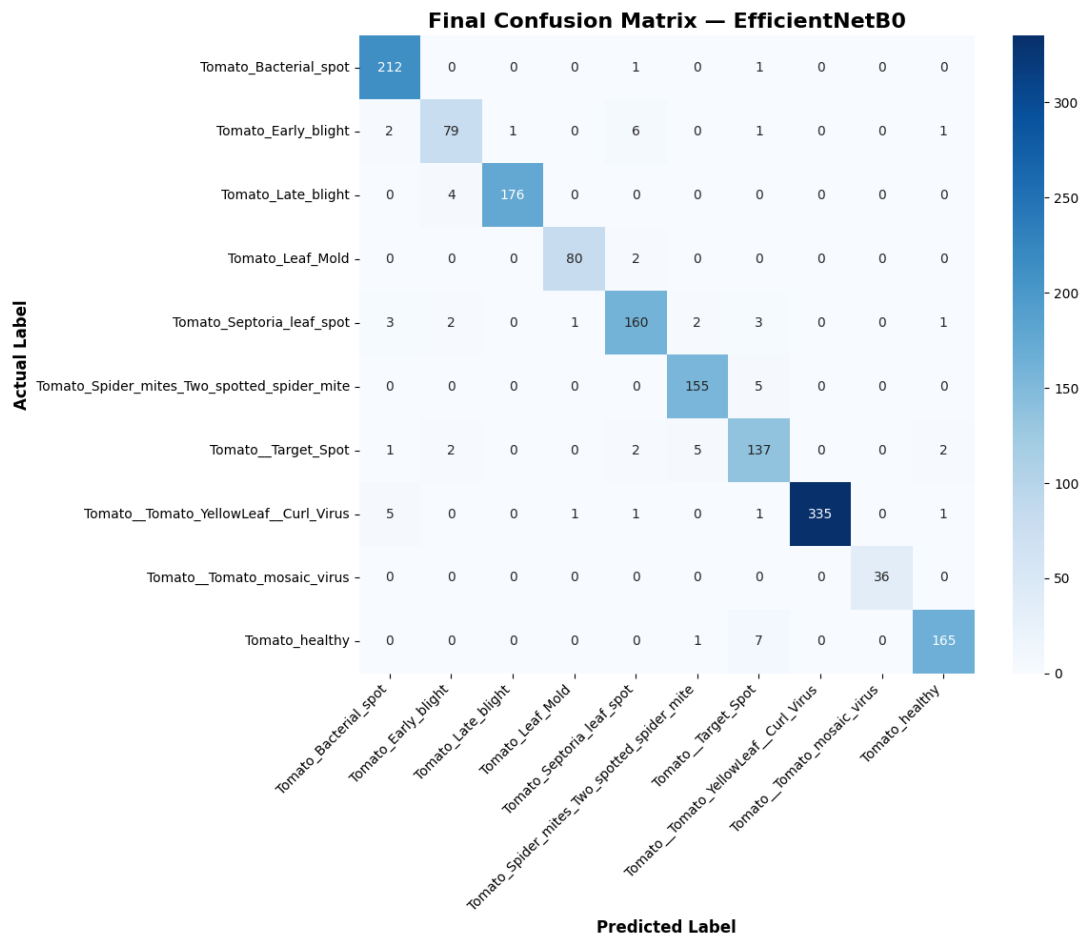
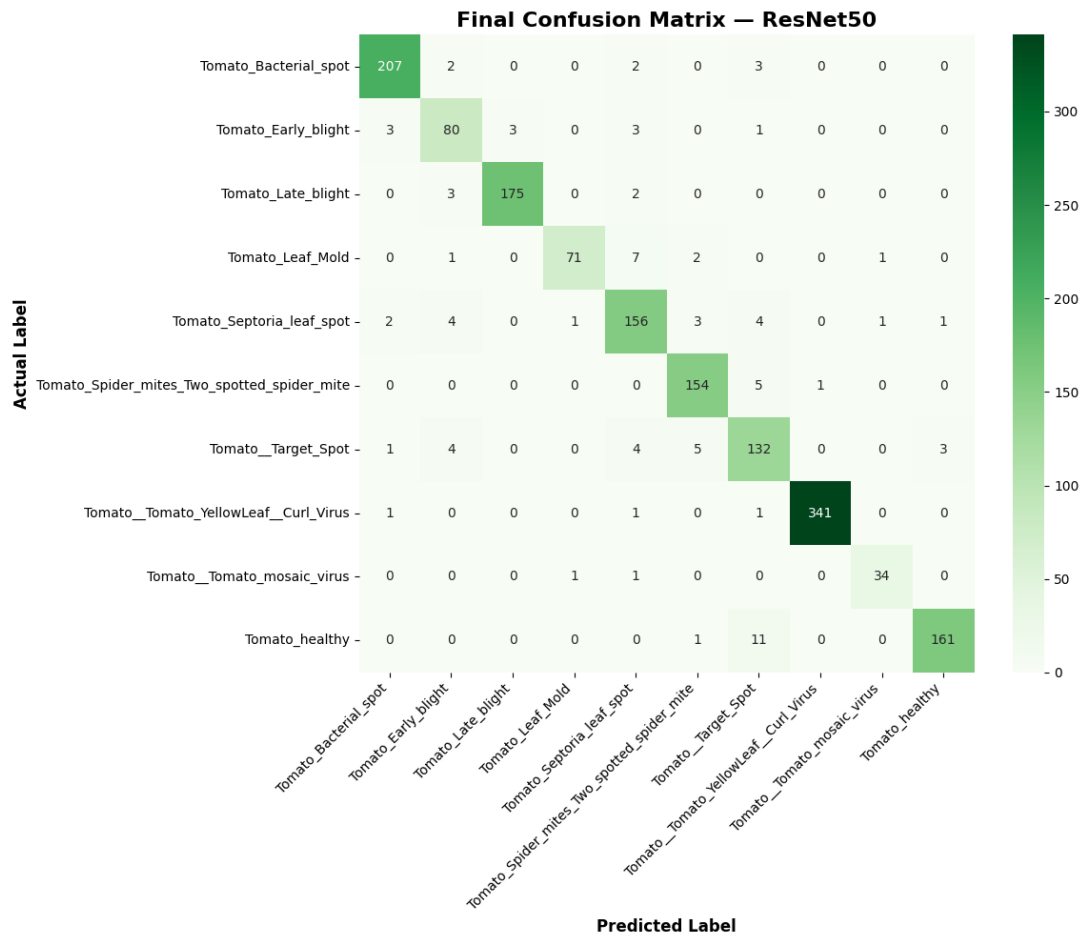


Figure 8: Comparative Confusion Matrices on the test dataset for the fine-tuned ResNet50 (top) and EfficientNetB0 (bottom) architectures.

8 Conclusion

This study presented a highly optimized, end-to-end machine learning framework for the automated detection and classification of tomato plant diseases. Recognizing that convolutional model performance is intrinsically linked to input data fidelity, a rigorous pre-processing methodology was established. By utilizing U-2-Net for high-speed salient leaf segmentation and applying Contrast Limited Adaptive Histogram Equalization (CLAHE) specifically to the luminance channel, the visual morphology of phytopathological lesions was significantly enhanced while strictly isolating confounding background variables.

Through a controlled comparative analysis of MobileNetV2, ResNet50, and EfficientNetB0, this research evaluated architectures based on both diagnostic precision and parametric efficiency. Following targeted unfreezing and fine-tuning governed by a micro-learning rate scheduler, EfficientNetB0 emerged as the superior network. Evaluated against an isolated test dataset, the model achieved a 95.94% overall accuracy, surpassing the 94.44% achieved by the significantly heavier ResNet50 architecture. Notably, it demonstrated near-perfect precision across multiple severe pathogen classes, proving its capability to distinguish complex, visually analogous disease states.

The integration of GPU-accelerated background segmentation, mathematical texture enhancement, and a lightweight convolutional architecture provides a highly scalable and robust solution for modern computational agriculture. The resulting framework is exceptionally well-suited for integration into continuous-monitoring environments, offering a high-accuracy, low-latency diagnostic mechanism critical for maintaining agricultural yield security.